

Building a Quiz Service with Spring Boot

This DIY project demonstrates the use of Spring Boot to develop a simple quiz service. The service exposes a REST API and a set of random questions, and generates a quiz with a set of ten questions.



Spring Boot is a Java framework for developing enterprise applications. With several sub-frameworks like Spring Data, Spring Rest, etc, Spring Boot helps in the quick development of full-stack applications. This DIY project demonstrates developing the server side of a Quiz application using Spring Boot.

The objective of the project is to develop the server side of a simple quiz application and expose it through REST API.

The API requirements are:

- An API to upload quiz questions. Each question consists of four options with one of them being the correct option. Each question belongs to a specific topic under a specific subject.
- An API to generate a quiz with a set of ten random questions on a specific subject and topic.
- An API to submit answers and get the score.

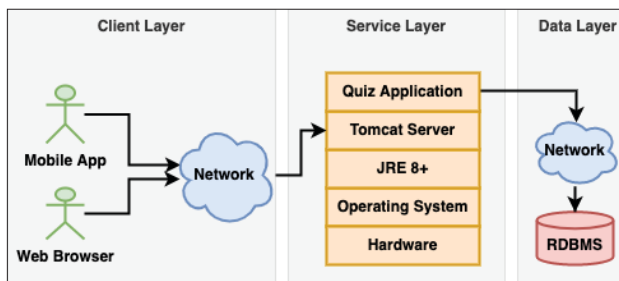


Figure 1: Architecture

The architectural requirements are as follows.

Database: Any RDBMS like MySQL

Environment: Spring Boot framework on Java 8+ platform with an embedded Tomcat server

The Quiz application represents a typical three-layer Java web application. The scope of this project is limited only to the service and data layers.

REST API

The Quiz service exposes the following three REST endpoints.

1. *POST /question* adds a new question to the system. Successful addition of the question returns the HTTP 201 status. The request payload will be a JSON with the following model:

```
{
  description: string,
  optionOne: string,
  optionTwo: string,
  optionThree: string,
  optionFour: string,
  answer: int,
  subject: string,
  topic: string
}
```

The response payload will be a JSON with the following model:

```
{
  qid: int,
  description: string,
  optionOne: string,
  optionTwo: string,
  optionThree: string,
  optionFour: string
}
```

2. *GET /questions?subject=SUBJECT&topic=TOPIC* gets a set of ten random questions from the specified SUBJECT